Forum:	Economic and Social Council (ECOSOC)
Issue:	Promoting efficient management of water resources in regions facing scarcity
Author:	Semeli Soulioti
Position:	Deputy President

Personal Introduction

My name is Semeli Soulioti and I am attending Year 9 at the German School of Athens. This year I have the utmost honour of serving as a Deputy President of the ECOSOC committee in this year's Moraitis Model United Nations conference. Through numerous debates and collaboration, I have learned that Model United Nations is not solely about the debate, but about raising young people's awareness of global affairs and reaching solutions through compromise and teamwork.

In this conference, I hope to work with all of you and create a welcoming environment. I truly believe that every delegate should feel comfortable, heard and not overlooked regardless of their experience or abilities. I hope to make everyone feel valued and ensure that every delegate has a chance to participate and express their opinion, seeing as I was once in their position.

The third topic of this year's ECOSOC agenda centres around the promotion of water management of water resources in areas experiencing water scarcity. Water scarcity is an insidious threat. It is a slow-moving crisis and it is detrimental to our ecosystems and economies. Thus, it requires efficient water management in order to mitigate these impacts. Apart from the study guide, I encourage you all to conduct research of your own to be able to understand the topic thoroughly.

I look forward to seeing you all in October! If you have any questions, please contact me via email at semsoulioti@icloud.com.

Introduction

Water scarcity and poor water management have had detrimental effects on society. To understand the importance of water, we have to first acknowledge water as an essential substance and a human right. Water is a limited and unevenly distributed resource as many individuals suggest, and it must be used responsibly as it is vital for our survival. Undoubtedly, water is taken for granted, disregarded and wasted despite it being the most vital substance on Earth.

Water scarcity has severe consequences threatening food security, our environment and social equality, while also endangering species. Efficient water resource management helps us diminish waste, enhance equal distribution and make sure that water resources are used responsibly. Without equal access to water resources, social conflict, health crises and forced migration can create life-

threatening conditions. Furthermore, the management of water resources should include vulnerable, communities with lower levels of literacy or numeracy, or less access to education in order to be accessible to all.

The problem is intensified by climate change, poor water governance and water pollution, deepening inequality. Water scarcity is not solely a lack of water but also about inequitable access to resources and institutions failing to balance the public's needs.

Key Terms and Definitions

Water stress

"Water stress occurs when the demand for water exceeds the available amount during a certain period or when poor quality restricts its use. Water stress causes deterioration of fresh water resources in terms of quantity (aquifer over-exploitation, dry rivers, etc.) and quality (eutrophication, organic matter pollution, saline intrusion, etc.)"[1]

Water scarcity

"Water scarcity is when there is not enough water, of sufficient quality, to meet the demands of humans and ecology. It occurs when the balance between supply and demand gets disrupted. A reduction in water supply, due to long periods of dry weather, can mean there's not enough water to meet water needs in an area."[2]

Economic water scarcity

"Economic water scarcity is due to a lack of water infrastructure in general or to the poor management of water resources where infrastructure is in place."[3]

Physical water scarcity

"Physical water scarcity refers to the situation where there are insufficient water resources to meet the basic needs of a given region. This is due to geographical or climatic conditions. Factors such as limited rainfall, high evaporation rates, and dry climates contribute to physical water scarcity. In

[1]Water stress. (2015). In *United Nations Economic and Social Commission for Western Asia*. https://archive.unescwa.org/water-stress

[2] What is water scarcity? | Beta | SEPA | scottish environment protection agency. (2018). In *Sepa.scot*. https://beta.sepa.scot/water-scarcity/what-is-water-scarcity/

[3] Water scarcity | description, mechanisms, effects, & solutions | britannica. (n.d.). In *www.britannica.com*. Retrieved July 23, 2026, from https://www.britannica.com/topic/water-scarcity#ref1265085

these areas, there is simply not enough water available to meet the needs of the population, agriculture, and industry."[4]

Irrigation

"Irrigation is the practice of supplying land with water so that crops and plants will grow."[5]

Aquifers

"An aquifer is a body of porous rock or sediment saturated with groundwater. Groundwater enters an aquifer as precipitation seeps through the soil. It can move through the aquifer and resurface through springs and wells."[6]

Precipitation

"Precipitation is water released from clouds in the form of rain, freezing rain, sleet, snow, or hail. Precipitation is the main way atmospheric water returns to the surface of the Earth. Most precipitation falls as rain."[7]

Background Information

Historical background

Understanding today's water resource management and water scarcity requires attention to how societies before us managed water over time. The roots date back to ancient civilisations developing early water collection and irrigation systems.

Water management systems date back to antiquity, when ancient civilisations, especially in the Mediterranean had limited and inadequate water resources due to the hot and arid climate. Due to these conditions the Ancient Greeks built hydraulic networks and constructed water storage systems, developing aqueducts, siphons, plumbing systems, cisterns etc. In Ancient Greece, cisterns, wells and fountains first appeared in the early Minoan period alongside other technologies. At the palace of Knossos the Minoans engineered plumbing systems supplying the palace with clean and flowing water as well as draining the remaining wastewater. The water supply was derived from springs

[4] anais. (2024). Ekopak - physical and economic water scarcity. In *Ekopak*. https://ekopakwater.com/en/physical-and-economic-water-scarcity-whats-the-difference/

[5] Irrigation | meaning in the cambridge english dictionary. (n.d.-a). In *dictionary.cambridge.org*. Retrieved July 23, 2026, from https://dictionary.cambridge.org/dictionary/english/irrigation

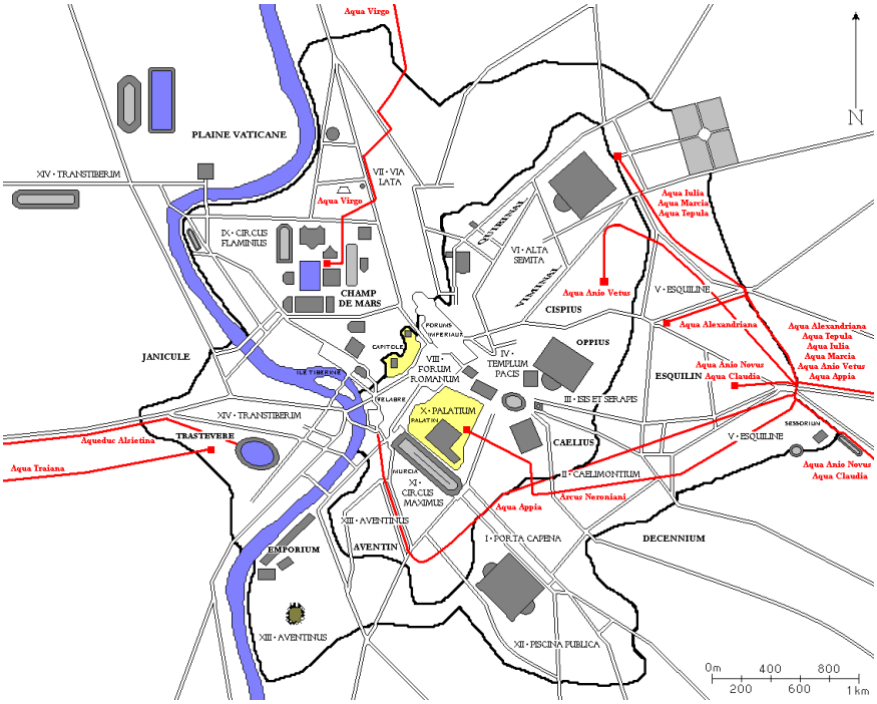
[6] National Geographic Society. (2022). Aquifers. In *education.nationalgeographic.org*. National Geographic. https://education.nationalgeographic.org/resource/aquifers/

[7] Precipitation and the water cycle. (2019). In *USGS*. https://www.usgs.gov/water-science-school/science/precipitation-and-water-cycle

using long pipes connected to the palace. The Minoans in Crete also constructed rainwater or precipitation collection systems which led water into large cisterns. One of the most impressive hydraulic works of the Ancient Greek era was the tunnel of Eupalinos on the island of Samos. The tunnel is a section of an aqueduct and is over 1,000 metres long. It was constructed to serve as an aqueduct and deliver freshwater to the city of Samos from external springs. It is estimated that the tunnel took around 8 to 15 years to complete and was used for over a millennium.

Apart from the Greeks, the Romans constructed immense gravity aqueducts, adequate underground water pipes supplying fountains, public baths, and sanitation with freshwater. In Mesopotamia, climate conditions were quite the opposite of what we would have expected. The Sumerians faced floods and seasonal drought. To tackle the issue the Sumerians built canals, reservoirs and dykes allowing them to manage the stream and ensure that crops and vegetation received sufficient water.

Thus, the history of ancient civilisations was central to the ascent of society, paving the way for future developments in water resource management.



Map of the 11 aqueducts in Rome[8]

Causes of water scarcity

[8] Wikipedia Contributors. (2019). Roman aqueduct. In *Wikipedia*. Wikimedia Foundation. https://en.wikipedia.org/wiki/Roman_aqueduct

Water scarcity has become a frequent phenomenon, increasing in severity in many regions over the years. It has become a life-threatening crisis impacting millions of people's lives. The adverse impacts of water scarcity are brutal and pervasive.

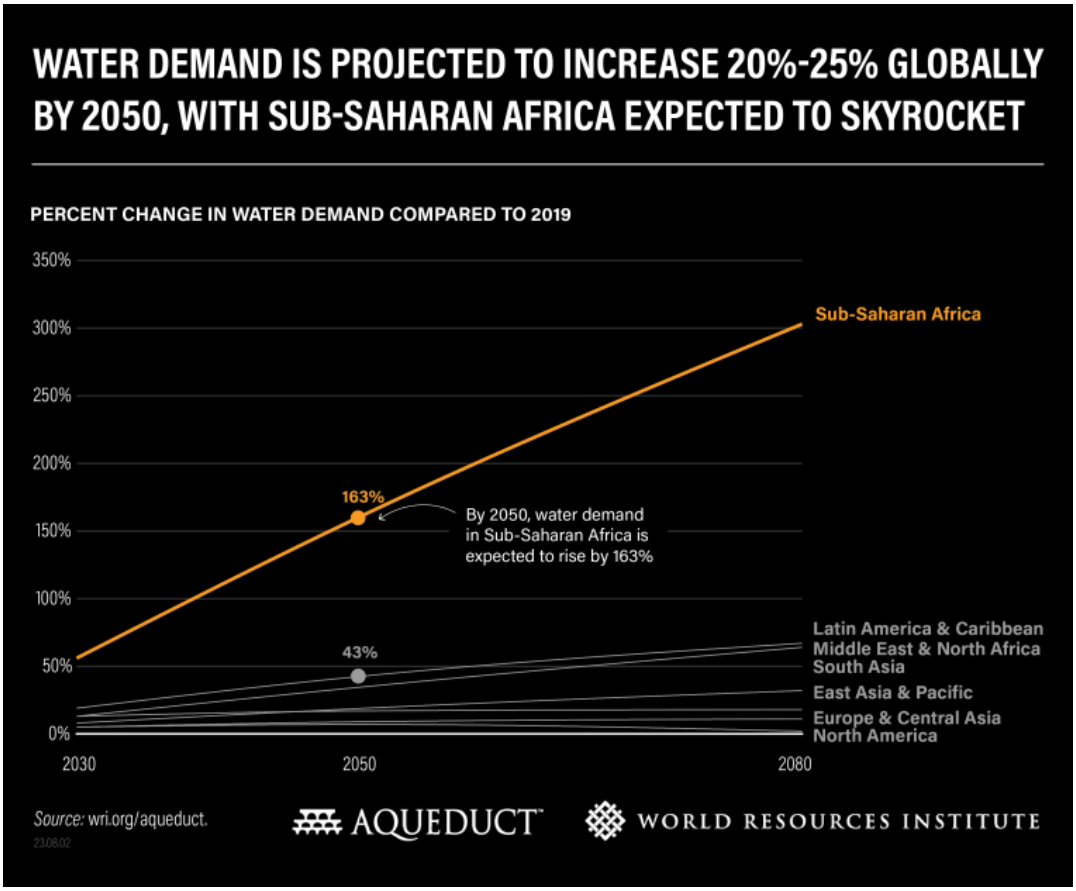
Climate change is one of the main causes of water scarcity as rising temperatures alter rainfall patterns, making water availability rather unpredictable, and sometimes bringing intense downpours. These extreme temperatures dry out soil and reduce surface water as water starts evaporating. Additionally, climate change causes extended periods of drought, possibly sparking fires which can reduce the soil's ability to absorb rainfall.

Agriculture is the most water-reliant branch of water consumption, and accounts for approximately 70% of global freshwater withdrawals. Due to water overuse and inefficient irrigation systems, many nations have reached their water resource limits. Excessive water usage causes chemical pollution through fertilisers and certain pesticides affecting both humans and animals. Often farms discharge their wastewater into large water bodies contaminating them with agrochemicals, jeopardising freshwater resources and aquatic life.

Furthermore, many areas rely solely on groundwater, especially when surface water is limited. When groundwater is pumped out faster than it is naturally restored water levels drop, causing water sources to shrivel. Pumping groundwater near the coast can cause saltwater intrusion, ultimately making the water saline. This happens when ocean water "intrudes" freshwater aquifers, occurring since freshwater normally sits on top of saltwater causing a natural barrier, though as this balance is disrupted saltwater rises. Salt water intrusion also affects agriculture, since soil structure degrades and disintegrates and reduces plant growth as they cannot survive off of saltwater.

Population growth is also a cause of water scarcity not because there are simply too many people but because there is growing water demand. Growing cities require more water for sanitation, public services and potable water for citizens. It also accompanies industrialisation, increasing water usage for cooling and manufacture. The danger of larger industries polluting aquifers or rivers has detrimental effects as it reduces accessible water. In brief, population growth means that more water resources are necessary, hence the demand for water usage in agriculture and households increases. Global freshwater demand is projected to increase through 2030.

In many regions the main causes of water scarcity are poor water management, inefficient irrigation systems or methods and weak water infrastructure. They are the more technical causes of water scarcity. Weak irrigation systems and methods lead to severe water wastage and inefficient, counterproductive use. Due to weak irrigation methods, substantial quantities of freshwater are used to produce food that is ultimately lost or wasted. Outdated facilities, flood irrigation methods, inadequate infrastructure and leaky pipes all contribute to water loss and ineffectiveness.



Projection of water demand globally by 2050.[9]

Impacts

Water scarcity has wide ranging effects on society and the environment. Water scarcity has already had major effects on households, ecosystems and development.

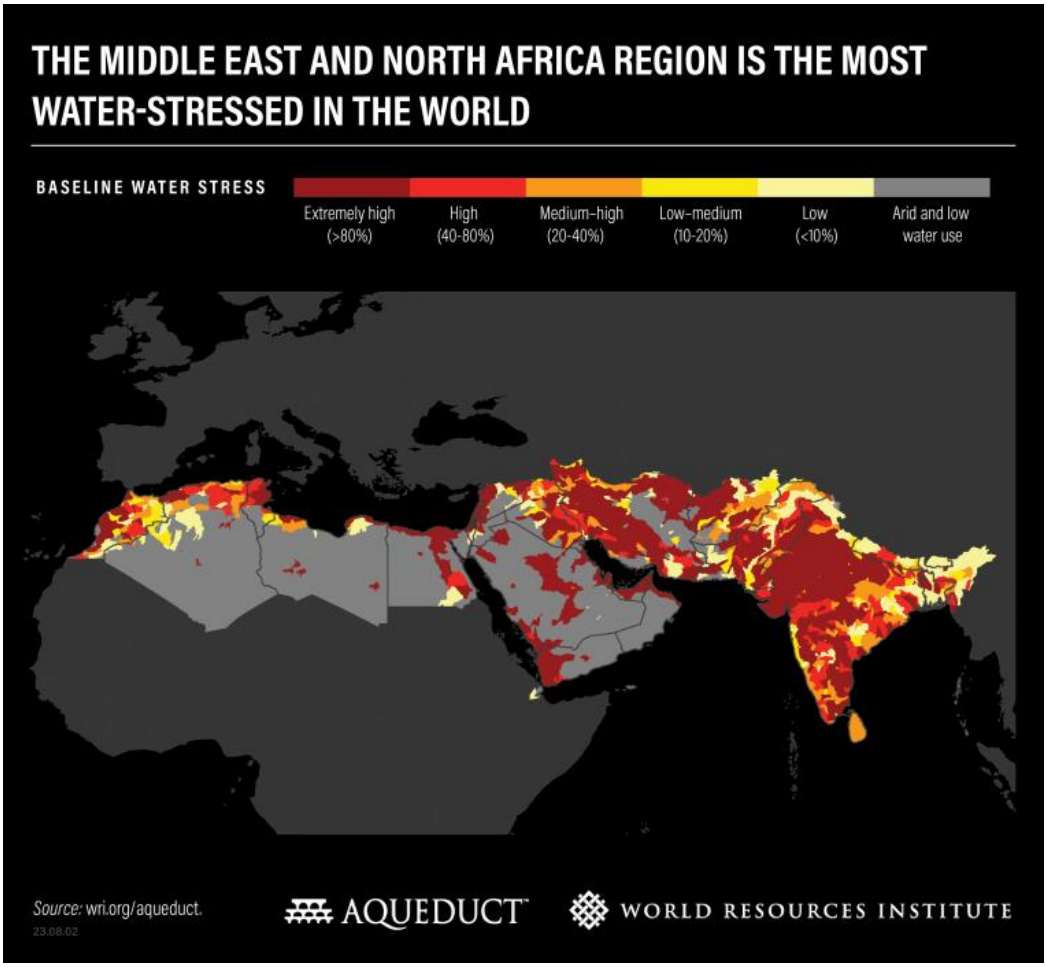
Without clean freshwater we cannot survive. Therefore, clean freshwater is essential for sanitation and human survival. However, around 2.1 billion people worldwide lack safely managed drinking water, while two-thirds of the global population experience severe water scarcity once a year. When water becomes unsafe, inaccessible or unavailable, people cannot drink enough, wash their food, while crops dry out and at worst the economy declines. Though the danger for humans doesn't stop here. It is estimated that 829,000 people die each year from diarrhoea attributable to inadequate water, sanitation and hygiene, according to an earlier WHO estimate. The majority of these deaths are in low- and middle-income countries. Additionally, malnutrition can be caused by poor sanitation through dirty water and harmful germs that trigger infections, damage gut health and the immune system, resulting in people not being able to absorb nutrients effectively.

[9] Kuzma, S., Saccoccia, L., & Chertock, M. (2023b). 25 Countries, housing one-quarter of the population, face extremely high water stress. *World Resources Institute*. https://www.wri.org/insights/highest-water-stressed-countries

The environmental impact of water scarcity is detrimental. Since 1970 freshwater species have been declining at an alarming rate of 83% as a result of habitat loss and pollution. As streams, rivers and lakes shrink, migratory fish species and wetland species decline. Wetlands support and house nearly 40% of all the world's plant and animal species making them a vital part of biodiversity. Wetlands keep vanishing slowly at a rate of 0.52% annually. Without wetlands ecosystems would collapse endangering thousands of species. Water scarcity, and habitat fragmentation are slowly pushing freshwater species and biodiversity towards extinction.

Middle East and North Africa (MENA)

The Middle East and North Africa (MENA) region is the most water-scarce region in the world. Countries with high water stress levels are located in the MENA region. The region's hot and arid climate as well as climate change contribute to water scarcity as they cause drought conditions and alter weather patterns. Though climate change is not the sole reason for water scarcity. Almost millions of people in the region lack adequate sanitation and hygiene services. Globally, agriculture accounts for 70% of freshwater withdrawals, while in the MENA agriculture accounts for a substantial share. Additionally, contributors to the area's high water stress levels are population growth due to migration caused by wars, poor water management and weak water infrastructure. Many countries in the MENA are facing socioeconomic challenges regarding water quality and water infrastructure making it harder to combat the issue of water scarcity. The demand for water has exceeded the region's capacity. As there is a growing demand for water due to limited freshwater supplies, women and children spend many hours just to collect water.



Map depicting the MENA region's water stress levels.[10]

Water related conflict

Water related conflicts are sparked by scarcity, control or weaponisation of freshwater resources. They can also involve deliberate attacks on water systems or water infrastructure.

During the US–Israeli military conflict with Iran, damage to Iranian water infrastructure has been reported. US and Israeli forces were accused of striking a desalination plant and disrupting the water supply on Qeshm Island in Iran. Damage to reservoirs was also reported. Taking into account that Iran was already facing a water drought and decline in precipitation, such damage could exacerbate an already serious water shortage.

[10] Kuzma, S., Saccoccia, L., & Chertock, M. (2023). 25 Countries, housing one-quarter of the population, face extremely high water stress. *World Resources Institute*. https://www.wri.org/insights/highest-water-stressed-countries

Throughout the Russia-Ukraine war, numerous attacks on Ukrainian civilian water infrastructure have been reported. In June 2023, the Kakhovka Dam was destroyed, flooding extensive areas of land and crippling irrigation of an agricultural hub.

AI and water

With the rise of artificial intelligence models in recent years, data centres have become prevalent in our modern world. It is safe to say that AI has impacted our society both positively and negatively. However, AI data centres have created a huge water consumption problem.

Generating a response requires electricity and electricity generates heat and therefore requires cooling. Some cooling systems use "evaporative cooling", which consumes water. The consumed water is generally just the water that evaporates during the cooling process. Many data centres use water-based cooling. The share of withdrawn water that data centres consume varies by cooling system. Data centres use different cooling methods which means their water consumption can vary considerably depending on the cooling method. During the evaporation process minerals and salts are left behind in the remaining water, sometimes damaging infrastructure, becoming corrosive and may require further treatment before reuse. This water is returned as wastewater, re-enters water circulation for other uses and can be reused.

One published estimate put the direct cooling-water consumption of training GPT-3 at approximately 700,000 litres. An earlier estimate suggested that generating roughly 10–50 responses could consume around 500 millilitres of water, depending on conditions, although if scaled to billions of queries worldwide, hundreds of millions of gallons of water are consumed. Data centres can range from medium-sized centres to massive facilities. A medium-sized data centre can consume up to 100 million gallons per year, while a *large* facility can use up to 1–5 million gallons per day. For comparison, some large facilities may use quantities of water comparable to a small town's daily consumption.

Furthermore, some data centres obtain cooling water from potable water supplies. Most of the water is drawn from blue sources, essentially freshwater resources such as aquifers, lakes and rivers. Data centres have multiple sources such as surface water and groundwater, purified and reclaimed water. However, companies are starting to change their cooling designs and methods, using different water sources such as greywater sources and processing it to a certain level. Greywater sources include reclaimed water from domestic use or wastewater.

Concerns on data centre water consumption are valid, and they merit attention. However, we must acknowledge the fact that yes data centres can affect the water supply in scarce regions but it depends on the location, cooling method and water sourcing as well as the data centre scale.

Timeline of Events

Date	Description of Event
23 October 2000	On the 23rd of October the Water Framework Directive was adopted by the European Parliament and the Council of the European Union. It was adopted to secure and restore water bodies.
28 July 2010	The UN resolution about the accessibility and the importance of water as a human right passed by the General Assembly.
2022-2023	Iran experienced a severe water crisis because of drought, low precipitation, and poor water management.
February 2026	In February of 2026, damage to Iranian water infrastructure was reported during the US–Israeli conflict with Iran.
7 March 2026	US forces were accused of striking a desalination plant on Qeshm Island in Iran.

Major Actors Involved

Iran

Since the early 2020s, the Islamic Republic of Iran has been experiencing a water crisis driven by droughts, aquifer depletion and high precipitation decline. Groundwater reserves are being overexploited leaving many aquifers under severe pressure. Their renewable water resources have declined by substantially, having led the country into acute water scarcity. Iran was pushed to the brink of water scarcity in part due to outdated policies and weak infrastructure. Iran was committed to agricultural sufficiency. They prioritised national food security over environmental conservation and sustainability. Their water infrastructure lacks proper oversight and their outdated water infrastructure contributes to leaks and intensifies shortages. In 2025 water supplies were cut in Tehran and experts warned people about water consumption during the day. Additionally, the water reservoirs in several major cities reportedly fell to very low levels. Nationally, several major dams reportedly reached critically low levels.

Afghanistan

Afghanistan's people face severe water shortages. The precipitation decline, climate change and the mismanagement of resources have left the country in severe conditions. Groundwater levels have declined. A large proportion of Afghans are affected by rising temperatures and droughts. In the countryside conditions are so harsh that residents depend on unsafe surface water for daily use, leading to waterborne illnesses. Only a minority of the population has access to *safely* managed drinking water. Due to their financial limitations and security risks, it is a challenge to resolve the issues within the water sector. Ever since the Taliban takeover, years of political instability and the COVID-19 pandemic, the country has struggled with underdeveloped water supplies and infrastructure. Rehabilitation efforts and international engagements have had mixed results.

Food and Agriculture Organization of the United Nations (FAO)

The Food and Agriculture Organization of the United Nations (FAO) is a specialised agency combating hunger and food insecurity among many other difficulties. The FAO cooperates with governments and international communities to also manage water efficiently, strengthening water rights, enhancing water governance of shared rivers. The FAO works to preserve, protect and allocate water resources in response to climate change and water scarcity. This organisation has established multiple frameworks on water such as *the Global Framework on Water Scarcity in Agriculture (WASAG).* Overall the Food and Agriculture Organization of the United Nations has made a great attempt in combatting the issue of water resource management providing solutions and help for people in need.

UN Documents and International Agreements

(A/RES/64/292)

On 28 July 2010, the UN General Assembly passed a significant resolution recognising "the right to safe and clean drinking water and sanitation as a human right that is essential for the full enjoyment of life and all human rights". It states the importance of water as a human right and its equitable accessibility. This resolution recognised access to safe drinking water and sanitation as a human right. They are related but remain separate rights. The key elements of both rights are availability and accessibility, both ensuring the access to sanitation and water facilities within quick reach and the access of them to all groups of people. Furthermore, some other elements would include quality and affordability, guaranteeing water services to all individuals and requiring that drinking water be safe and free from harmful levels of contaminants, which can cause health problems.

Water Framework Directive (WFD)

The Water Framework Directive was adopted by the European Parliament and the Council of the European Union on 23 October 2000. The WFD is a framework focusing on the protection and partly the restoration of water bodies. It concentrates on optimal wellbeing and diminishing pollution. This

directive ensures that water can support both wildlife and ecosystems, but concurrently human needs. The Water Framework Directive has been the main law for water conservation in Europe. Additionally, the WFD applies to groundwater and inland, transitional and coastal surface waters and is based on a water management strategy that uses a river basin as the core unit. This approach makes sure that surrounding countries can harmoniously cooperate to regulate rivers and common water bodies. Two of its key goals are to meet high ecological and chemical standards for surface water and groundwater and most importantly implement safety and security measures around water basins.

Previous Attempts to Solve the issue

The AWSAMe initiative

AWSAMe (Addressing Water Scarcity in Agriculture and the Environment) is a WASAG (the Global Framework on Water Scarcity in Agriculture) linked initiative led by the FAO. This initiative aims to promote drought and climate resilient crops, and efficiency in irrigation systems alongside enhancing output and water use efficiency. AWSAMe implements the modernisation through indigenous and prevalent reforms and practices. Furthermore, this initiative was implemented with the government of Malawi. In Malawi they dealt with a decline in precipitation as well as general water scarcity due to their climate, affecting their traditional crops. Farmers are highly susceptible to these climate conditions and their crops fail when rainfall is low. They face agricultural as well as nutritional issues, possibly threatening food security. Thus, they started promoting drought and climate resilient crops that can adapt to dry conditions reducing the farmers' dependence on water. AWSAMe project cooperated with local agricultural establishments and most importantly fostered sorghum and cowpea crops amongst rural communities and enhanced the distribution of these seeds, encouraging seed planting. In brief, this initiative has shown promising results and raised awareness and educated locals on water demand. Although, The evidence does not indicate or inform us on how much water was saved or whether the project reduced the general local water demand. Additionally, it is uncertain whether the solution actually lasted. The project has potential and can be expanded widely, but it is still being developed and evaluated.

Israel

In recent years, severe weather conditions and climate change have been more frequent in Israel. Israel is vulnerable to the impacts of climate change and its climate is already dry and arid. The projected population growth and the diminishing supply of water sources present a challenge for maintaining access to people. However, Israel has adapted to its arid climate and their insufficient water resources. They have reformed their water sector in order to provide their growing economy with adequate and reliable water, through regulatory, conventional reforms and infrastructure investment. Israel has managed to lessen overexploitation of their natural resource by reuse of treated wastewater and desalination of saline water. They have established desalination plants supplying over 80% of the country's municipal water. Additionally, their wastewater is treated up to a

favourable extent and reused for primarily agriculture. Overall, Israel has made substantial progress in the water sector and has improved water plans through technologies and advanced tools. They have efficiently tackled their water shortage issues through sustainable water management and set an excellent example for other countries to follow.

Possible Solutions

Investing in wastewater treatment and reuse

Investing in wastewater treatment and reuse is very beneficial for both water management and water supply. Wastewater typically refers to used urban or industrial water. Following collection and appropriate treatment, wastewater can be reused for agriculture and industry. However, treated wastewater can also be reused for drinking purposes where stringent safety standards and monitoring are met. In order for wastewater to be reused for agricultural or industrial purposes, it must first be treated which happens through multiple steps. This treatment requires chemical and biological methods for wastewater to be considered suitable for its intended use. During the process, contaminants and chemicals are removed, purifying water. Wastewater reuse reduces reliance on scarce freshwater supplies and establishes more reliable local resources. Wastewater reuse also offers other benefits. Using reclaimed water for agricultural irrigation improves water management sustainability and reduces wastewater. In agriculture, wastewater reuse will ensure a constant and consistent water flow even during periods of drought or severe climate conditions, essentially improving agriculture irrigation. Ultimately, this solution will lessen water waste and rather promote its recycling and reuse among different sectors.

Enhancing water infrastructure

In many countries around the world water infrastructure and facilities are insufficient or weak. Water infrastructure manages water resources such as their distribution and treatment processes. By water infrastructure we mean water pipes, wastewater treatment systems, water supply and storage systems in general. Water infrastructure is a key element to the sustainability of our environment and our human needs. Improving weak water infrastructure replaces previous failing components, reduce the risk of recurring service failures and generally make infrastructure and facilities more resilient. Given the financial constraints of low or middle income countries, external financing, public–private partnerships or development bank loans could facilitate governments and institutions in upgrading their infrastructure. Clearly, countries must reflect on their own problems regarding water infrastructure and make adjustments according to their system deficiencies. Improving their water infrastructure can become a reliable asset and become a firm foundation of human lives and ecological stability.

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